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Introduction

Objective

The project test whether VIX based regime can improve allocation tatically across 3 liquid ETFs SPY for US equities, TLT for long duration US treasury and GLD for gold exposure. Aim to improve return, improve risk adjusted performance and reduce drawdown with respect to benchmarks : equal weighted portfolio and S&P buy & hold.

Why are we doing this?

I selected 3 ETFs SPY, TLT and GLD variants of ETFs react differently across different market situation. There are various phase of periods which impacts the market return and behavior. For instance- during calm period when market is booming – normally stocks tend to perform quite well compared to the other products like bond. During normal period-i.e. stagflation- bonds or fixed income do better . However, when the market is very stressful like that of now(oil crisis, global supply chain disruption, covid in 2020) - precious commodities like Gold and government treasuries often give stable to better return. This motive drives the strategy which can detect the changing market scenarios and then accordingly change the allocation of each of these variants of asset class weights so to get higher risk-adjusted return and suffer minimal losses compared to benchmark of simply buying and holding stocks. The intelligent investment decisions provide better protection to client capital during turbulent times.

Main concepts used

In order to be able to demonstrate the prevailing market stress or volatility situation- we use the CBOE Volatility Index popularly refer as VIX Index and which is also tagged as important element of the market stress. Using the concepts from the our study package of Markov chains[1] and Hidden Markov Models[2]- we try to model and identify hidden “regimes” ie market models which we cannot observe directly and we can only inference via from the historical pattern study of how the VIX changes during the horizon. These probabilistic and stastical backtested models also help us to study and model the transitions phase among one market state to other ie allow us to determine when likely current regime at any given point of time

For each market models on regime (ie calm, normal & stressful)- we evaluate which out of the 3 ETFs has historically given us better return and perform well compared to others. We then devise simple strategy rule based model as : calm regime--> hold SPY, if normal/stress-->switch to TLT or mix of gold & TLT. We then test this rule based market model on real historical data(2005 – 2026) as realistic as possible by

- Adding trading delays

- Incurring transaction costs

Thus the combination give us asset selection & backtested results to turn statistical pattern in practical investment policy

Step 1: Data Preparation & Exploration

Before building strategy- we build need to able to clean, reliable data for the 3 ETFs and VIX index
We followed the following steps :

- Downloaded close adjusted prices for TLT, GLD , SPY and VIX from Y-Finance for the period common among the all – 2005 to 2026(latest date 15-May 2026).
- Each of these prices are then converted to daily log-returns for the ETF and daily changes for ΔVIX for the volatility index daily change.
- Any missing days are removed in order to ensure line up perfectly along the common date range.

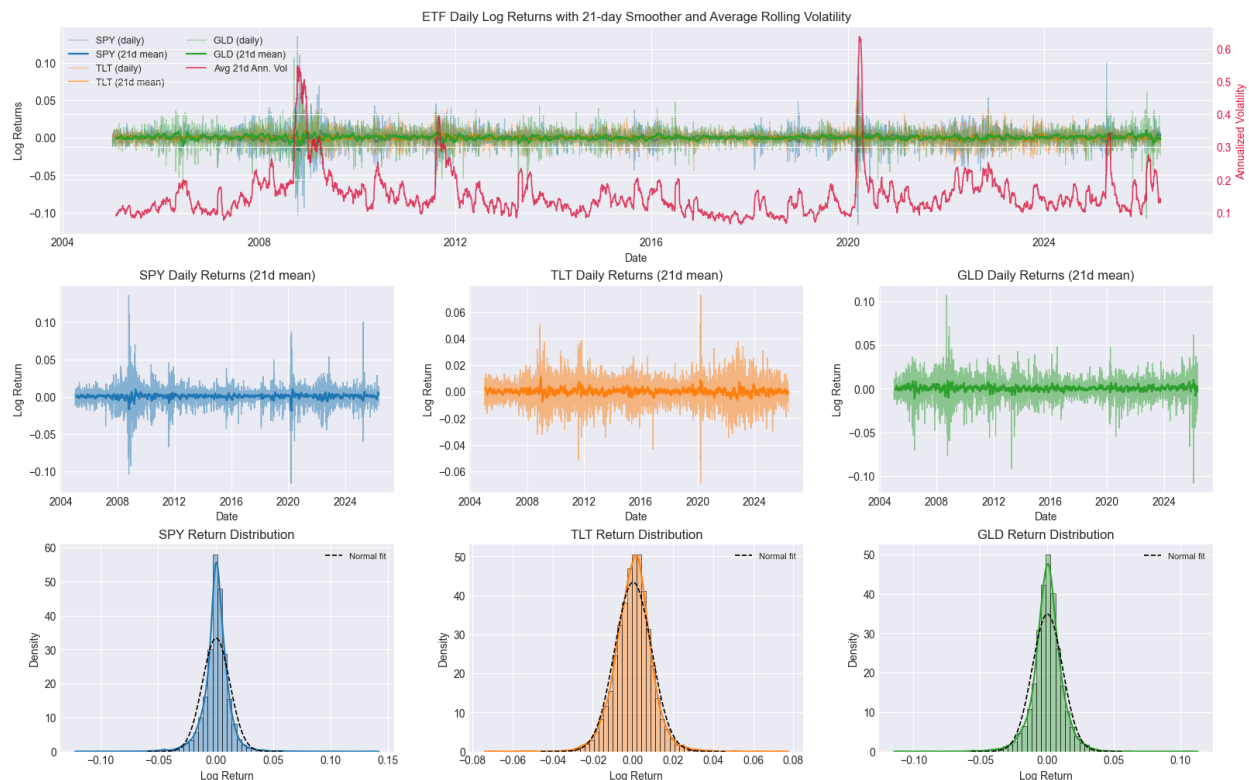


Fig1: ETF Daily Log Returns, Rolling Volatility, and Return Distribution Characteristics (2005–2026)

Historical ETFs returns :

- Above historical analysis shows historical returns of SPY, TLT and GLD over common period.
- It also depicts volatility clustering where during period of stress such as 2008 financial crisis and Covid-19 in 2020 where in the volatility increases due to unexpected scenario
- SPY among 3 shows maximum fluctuations in return and spikes- this is as per expectation as equity market more sensitive during economic shock and sentiment.
- TLT acts defensively- smaller spike or swing thus providing safer assets during stressful period.
- GLD appears independent to SPY & TLT thus helping with diversification benefits.
- The histogram at bottom shows about imperfect normal distribution ie having fatter tails and more extreme observations .

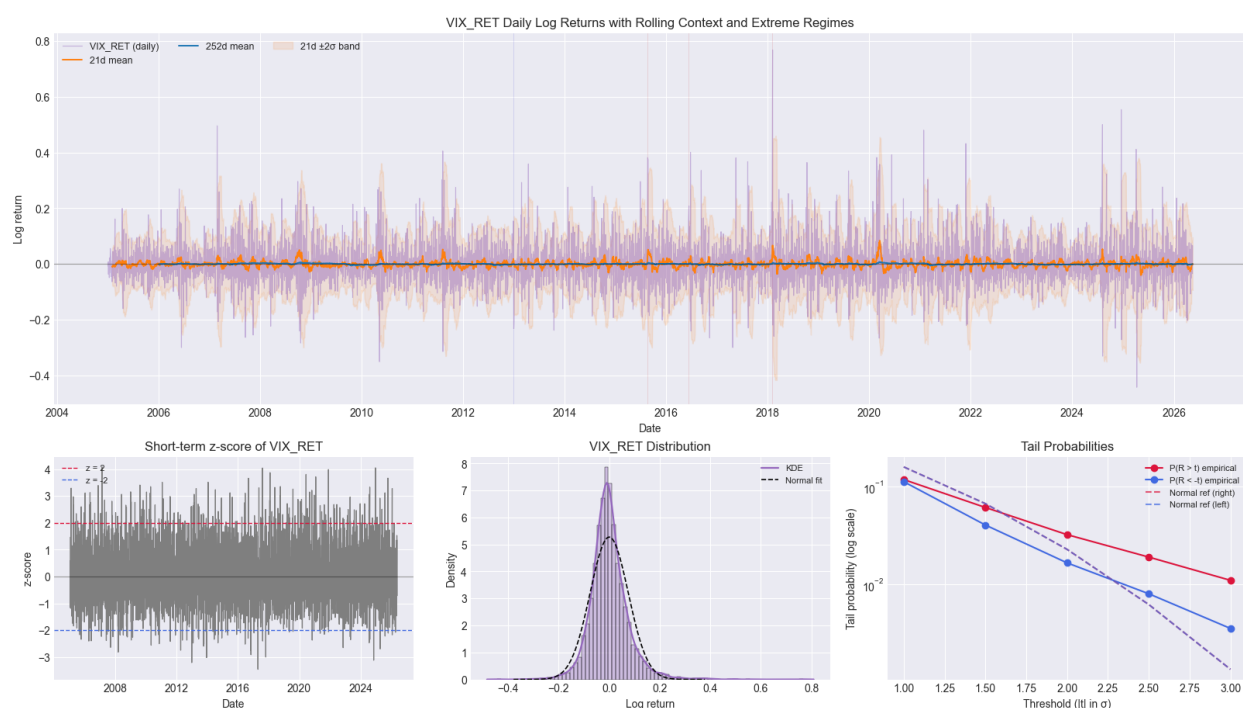


Fig2: Distributional Characteristics of VIX Daily Log Returns

VIX daily log returns:

- We can observe large movement in VIX index trend during the concentrated stress market and not evenly distributed. Eg 2008 financial crisis, Covid-19 market shock are periods where amplitude increased sharply.
- VIX is heteroskedastic ie not describable using constant variance assumption- this is quite intuitively depicted by rolling mean close to zero for most but changing across environments.
- The z-score provide explanation for deviations more frequently.
- The bottom histogram & tail-probability shows for Gaussian behavior separation. It has heavier tails and stronger asymmetry compared to fitted normal distribution

- Right tail is more pronounced as VIX tends to spike sharply during panic

Table 1: Summary Statistics of Daily Log Returns

Asset	Mean	Std Dev	Skewness	Kurtosis
SPY	0.000410	0.011962	-0.304	14.714
TLT	0.000118	0.009198	0.004	3.435
GLD	0.000423	0.011449	-0.454	6.861
VIX_RET	0.000050	0.075655	1.029	6.512

Summary statistics:

- TLT gives lower average but with lower vol
- GLD and SPY provides negative skewness suggesting that we will get larger downside more frequently than that of larger positive moves.
- Kurtosis more than 3 for all indicates fat tails and very extreme observations . SPY being highest as larger market shock impact equity more
- VIX returns have high standard deviation as per the table due to larger positive spike when market is in panic situation thus it is causing non-asymmetric distribution and also with heavy right tail.

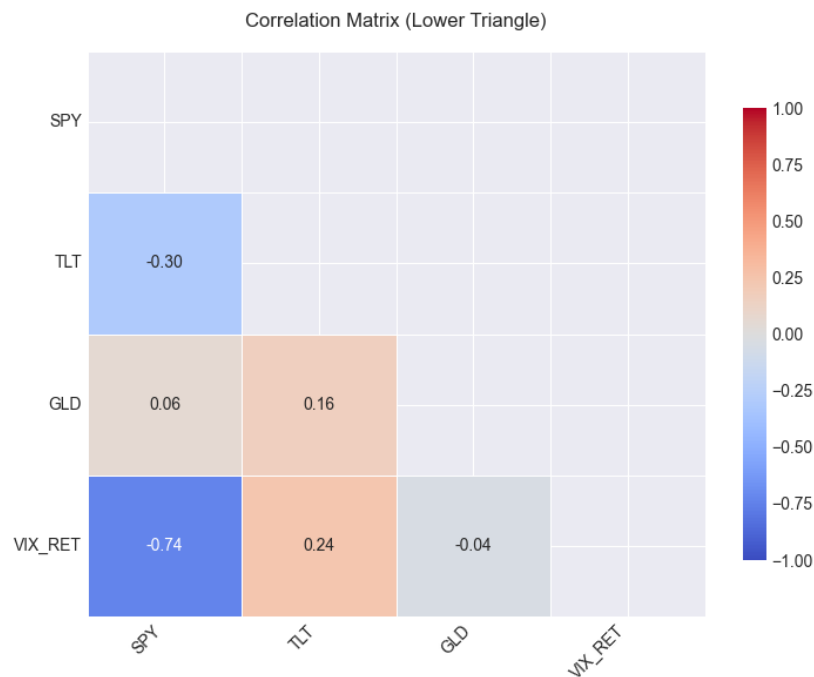


Fig3: Correlation among the 3 ETFs and with VIX_Return

Correlation statistics:

- SPY $\leftrightarrow$ VIX log-change = -0.7 shows strong negative correlation confirms that VIX is contemporary to equity fear gauge
- TLT is mildly correlated while GLD largely uncorrelated with both SPY & TLT- providing justification for diversifier

Step 2: Modeling VIX Regimes

Markets change from calm to stress and hence we need systematic way to detect which period is currently the market is and so that one can adjust the portfolio to reflect the right action as per prevailing period

We proceed with following approach:

- We tested two approaches for this 1. Discrete Markov Chain option and 2. Hidden Markov Model option
- Discrete Markov Chain option[3]
 - We took the Δ VIX series data and discretized into 2 or 3 simple market states ie low/medium/high volatility using the quantile thresholds
 - Then we created a transition matrix – providing us information about the probability of moving from 1 state of today's market to another state of tomorrow's market mood.
 - Also calculated the stationary distribution to understand long run proportion of time market spends in each of the given state
 - The stationary probability calculated comes to be $1/3^{\text{rd}}$ or 33% suggesting states are not very sticky
 - It is simple and transparent but do have mechanical in nature as its contain roughly $1/3^{\text{rd}}$ observation
- Hidden Markov Model option (HMM)
 - HMM helps with improved regime interpretation.
 - we took help of Gaussian Hidden Markov Model to fit 2-state and 3-state to the same Δ VIX series data
 - The fitted HMM parameter help to show that three regimes differ in their statistical parameters like their mean and their variance of VIX returns.
 - High volatile or stress regime has higher positive VIX-return while lower volatile(calm) has lower.
 - It also helps with posterior probabilities which is kind give idea about the confidence of model for its state for each day
 - This data then we trained using EM (Expectation-Maximization)[4] algorithm where we update the parameter iterative way to explain observed change in VIX.
 - Finally, taking help of Hamilton forward filter, we computed the hidden regime at each of the day and also smoothed probabilities to show confidence of each regime.
 - Our manually computed log-likelihood matches with fitted HMM log likelihood as shown using Hamilton filter verification step

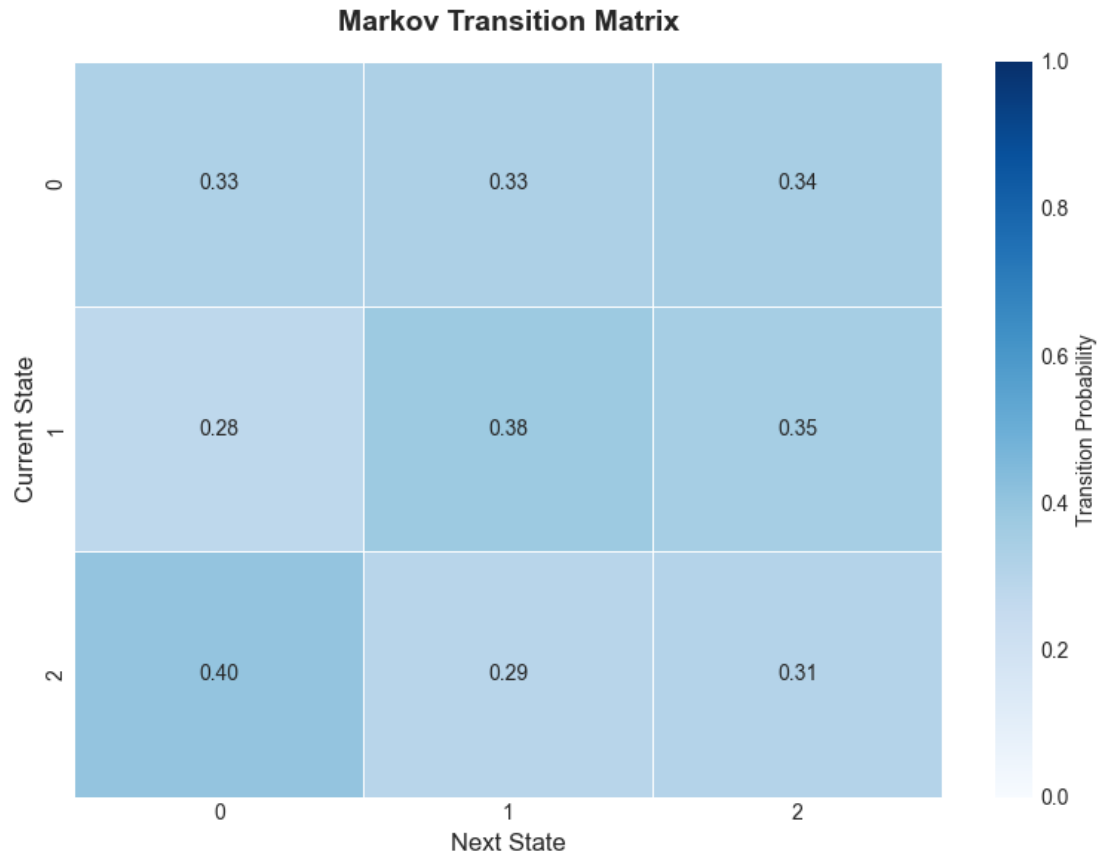


Fig4: MC-3 transition matrix show probabilities moving between states 0,1,2

Table 2: MC-3 Stationary distribution

State	P(stay)	E[duration] (Days)	Stationary π
State 0	0.325	1.482	0.333
State 1	0.375	1.601	0.333
State 2	0.307	1.442	0.333

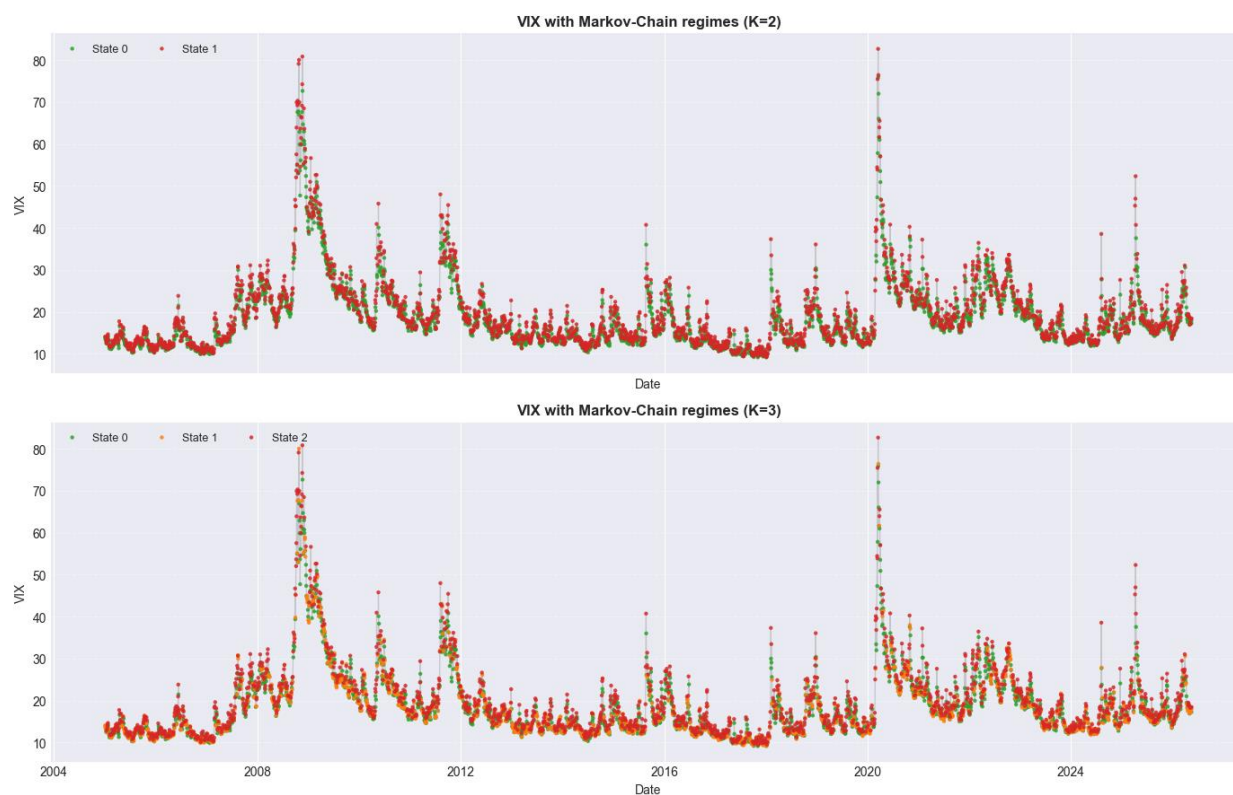


Fig5: VIX with MC states ($k=2$ & $k=3$)

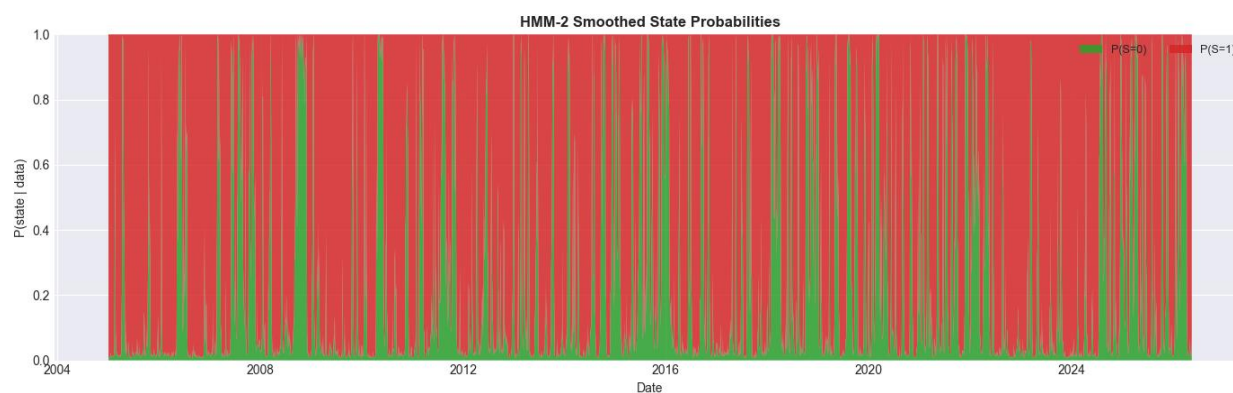


Fig6: HMM-2 states fitted State probabilities

Table 3: HMM-2 State fitted Parameters (Means, Variances, and Initial Probabilities)

State	Mean	Variance	Standard Deviation (σ)	Start Probability (π_1)
State 0	0.01608	0.015525	0.12460	0.0
State 1	-0.00526	0.002365	0.04863	1.0

Table 4: HMM-2 Transition Probability Matrix

From / To	State 0	State 1
State 0	0.847	0.153
State 1	0.051	0.949

Table 5: HMM-3 State fitted Parameters (Means, Variances, and Initial Probabilities)

State	Mean	Variance	Standard Deviation (σ)	Start Probability (π_1)
State 0	0.05765	0.003243	0.05695	0.0
State 1	-0.01429	0.001843	0.04293	1.0
State 2	0.00857	0.017131	0.13089	0.0

Table 6: HMM-3 Transition Probability Matrix

From / To	State 0	State 1	State 2
State 0	0.031	0.771	0.199
State 1	0.197	0.803	0.000
State 2	0.000	0.132	0.868

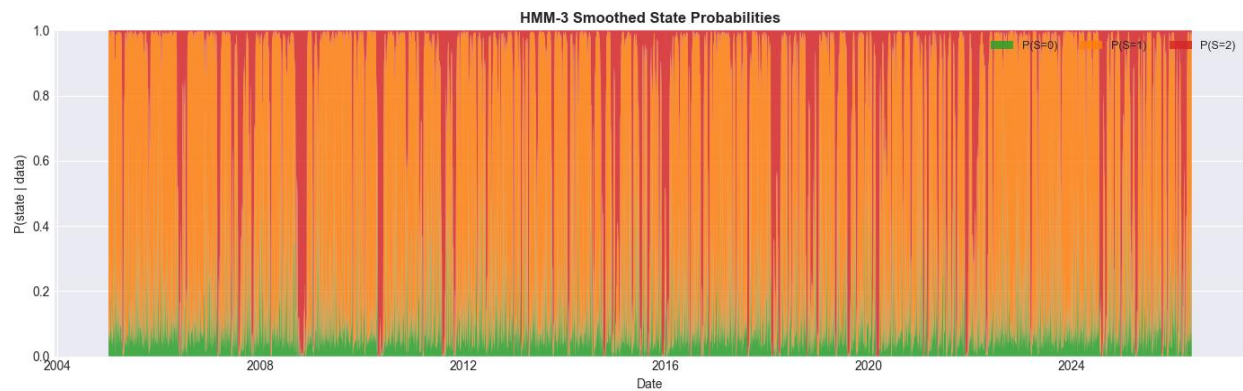


Fig7: HMM-3 states fitted State probabilities

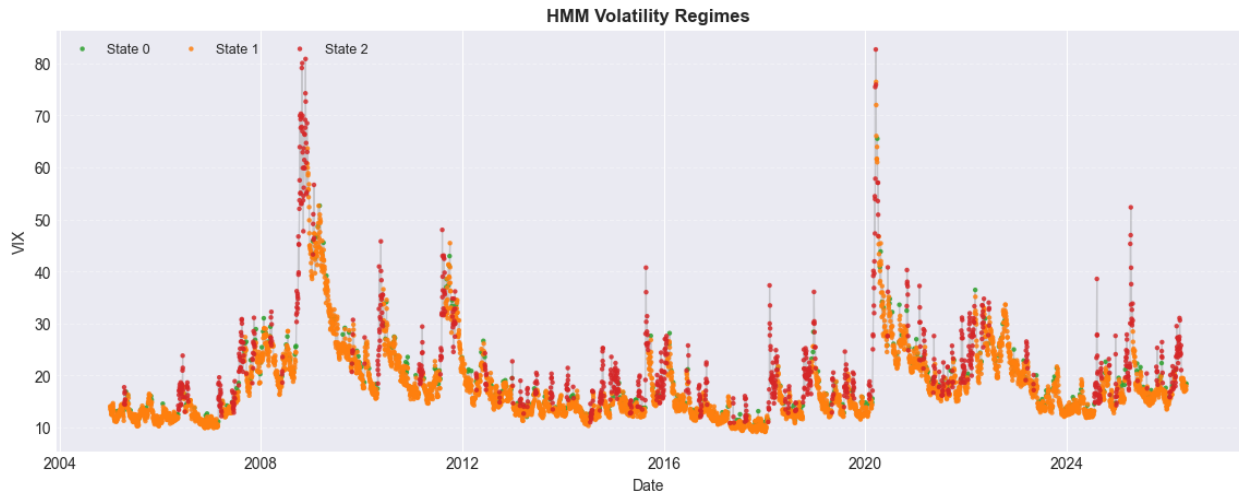


Fig8: VIX with HMM-3 color-coded states

Inferences from the above Outputs

1. Markov Chain transition matrix

- As states are created using quantile cut-offs- we observed almost balanced among low, medium & high VIX return regime
- Also- shows switching is quite frequent which implies states to be noisy and less stable than HMM

2. Markov stationary distribution

- Shows long-run probabilities and equal across states
- This deviates from real markets, as may not capture crisis periods

3. HMM fitted parameters

- More informative as it allows for estimating hidden regimes
- Clear demarcation of stress regime – via high positive VIX return mean viz a viz calm regime with negative VIX return mean

4. HMM transition matrix

- It shows persistence as some regime has high probabilities .
- It also signifies that volatility regimes are not random ie once the market enters a particular regime – it may remain there for sometime

5. VIX state plot

- It shows that model identifies stress regimes when VIX rises sharply

Overall-both MC & HMM methods can classify VIX behaviour in regimes. MC approach is very simple but dependent on chosen quantile. HMM turns to be flexible as it learns from hidden regimes and thus gives stronger & economically viable representation of regimes.

Step 3: State Selection and Interpretation

Context: to choose best regime candidate model using various comparison techniques

Process:

- We compare MC-2, MC-3, HMM-2 and HMM-3 using log-likelihood, AIC & BIC
- Then study how each ETFs type ie SPY, TLT and GLD perform in each selected state.

Table 7: Model Comparison Table

Model	K (parameters)	Log-Likelihood	AIC	BIC
MC-2	7	4,520.14	-9,026.27	-8,980.15
MC-3	14	4,842.49	-9,656.97	-9,564.72
HMM-2	7	6,879.58	-13,745.16	-13,699.03
HMM-3	14	6,941.35	-13,854.70	-13,762.45

Inference:

- HMM-3 is preferred model .
- It has highest log-likelihood, lowest AIC and BIC among all . It implies even if HMM-3 state have more parameters- improvement in model fit large enough to justify complexity

Table 8: HMM-3 selection justification

Criterion	Reasoning
BIC / AIC	HMM-3 attains the lowest BIC and AIC in the comparison table above
Interpretability	states map cleanly to <i>calm</i> , <i>normal</i> , and <i>stress</i> volatility regimes
Stability	probabilities (plot above) show regimes are stable and hence allows for signals are tradable & Economical.
Visual plausibility	The 3 color-coded VIX aligns the crisis episodes (2008, 2011, 2015, 2020, 2022).
Vs. MC	The quantile Markov Chains ignores Gaussian emission structure; HMM uses the full distribution and yields a good likelihood per parameter.

Table 9: HMM-3 state conditional return & volatility table

HMM_3_STATE	SPY Mean	TLT Mean	GLD Mean	SPY Std Dev	TLT Std Dev	GLD Std Dev
0	-0.01082	0.00332	-0.00065	0.00859	0.00974	0.01229
1	0.00231	-0.00042	0.00060	0.00803	0.00843	0.01042
2	-0.00195	0.00078	0.00020	0.01988	0.01121	0.01437

Inference of state conditional return & standard deviation:

- It shows SPY has negative while TLT has positive average return. This suggests – that TLT are useful during volatility spike.
- On the other hand- in calm state SPY has strongest average return and so equities are very much chosen
- In normal state- TLT has highest average return hence strategy assign state to TLT

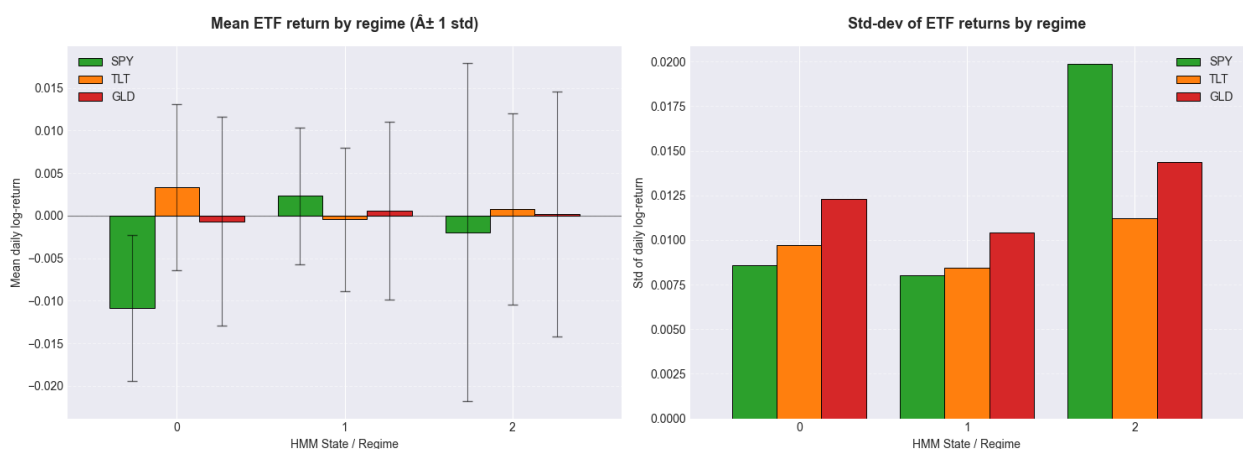


Fig9: Bar chart of state-conditional returns & standard deviation

Step 4: Designing the rotation strategy

The purpose of this step to convert the regime model into economical investment rule. HMM gives the market state but portfolio still need a decision rule to tell which asset to hold in each state. Idea is to build simple, transparent & easy to follow strategy.

Here we proceeded with below steps:

- Using the recommended HMM-3 regime sequence – we calculate the average return of SPY, TLT & GLD in regime situation.
- Then for each state – we try to choose combination ETF with the highest average return
- 1-day execution lag is used as well to avoid look-ahead bias. This to ensure strategy doesn't tomorrow's return to decide to decide today's position. Regime signal from day $t-1$ is used to hold asset on day t .
- Base rule based rotation strategy ie top-1 rule

Choose ETF = $\arg \max E[r | \text{state}]$

- Markov Decision Process interpretation
 - This gives theoretical explanation for simple rule based strategy

Component	Description
States (S)	$\{0, 1, 2\}$ — the inferred HMM-3 volatility regimes
Actions (A)	SPY, TLT, GLD — select one ETF for the next trading day
Transition Probabilities	P of moving from 1 state to another
Reward Function	Average ETF return each state
Discount Factor (gamma)	0.99

Table 10: HMM-3 allocation mapping table

State	Regime	Mean ΔVIX	Std ΔVIX	Chosen ETF	w_SPY	w_TLT	w_GLD	Rationale
0	Stress (VIX spikes)	0.09807	0.03443	TLT	0.0	1.0	0.0	Highest mean return in this regime (TLT)
1	Calm (low ΔVIX)	-0.01379	0.04094	SPY	1.0	0.0	0.0	Highest mean return in this regime (SPY)
2	Normal	0.01035	0.13398	TLT	0.0	1.0	0.0	Highest mean return in this regime (TLT)

The final rule:

- If regime is Stress-> allocate 100% to TLT
- If calm->allocation 100% to SPY
- If normal-> allocate 100% to TLT

Table 11: Markov decision process Output

State	Regime	MDP Optimal Policy	Top-1 Heuristic	$V^*(s)$	$Q(s, SPY)$	$Q(s, TLT)$	$Q(s, GLD)$
0	Stress (VIX spikes)	TLT	TLT	0.2151	0.2009	0.2168	0.2046
1	Calm (low Delta VIX)	SPY	SPY	0.2158	0.2141	0.2131	0.2059
2	Normal	TLT	TLT	0.2061	0.2098	0.2143	0.2055

Inference:

- MDP[5] solution converges in 2 steps and it matches top-1 rule.
- This signifies MDP is also theoretical confirmation that simple state to asset rule is consistent with Bellman decision framework
- Under the simplified MDP assumptions, the policy matched top-1 rule

Optional Strategy Work:

- We also considered optional extension.
- We considered 60/40 top two strategy.

- We put 60% in best ETF and 40% in 2nd best ETF for the regime. This allows us to reduce concentration risk
- Transaction cost- strategy switches too often look good but weaker after cost. We uses transaction coset netting in backtest to make evaluation realistic.
- Q-learning- we use as reinforcement learning check. It learns value of each action from repeated experience.
- Q learning model converges ,recover and show close Q value for TLT & GLD. This shows defensive asset work during stress periods.

Step 5: Back testing and evaluation

After building the strategy – we are going to backtest it over common horizon time period using the selected HMM-3 regime signal . Additional we also insert 1-day lag which allows to improve or avoid forward looking biasness.

The cumulative log return performance we calculated for the strategy is then compared with our 2 standard benchmarks ie Equal weight benchmark and another one SPY Buy & Hold benchmark.

We have presented our comparison of the performance in the following table.

Table 12: Performance Summary Table

Metric	Strategy	Equal Weight (Monthly)	SPY Buy and Hold
Cumulative Return	4910% (50.1x)	460% (5.6x)	712% (8.1x)
CAGR	20.3%	9.3%	10.9%
Volatility	15.1%	9.7%	19.0%
Sharpe Ratio	1.35	0.95	0.57
Max Drawdown	-26.0%	-22.9%	-55.2%

We observe that our strategy likely produces 50.1 times, compared with 5.6x and 8.1x for equal weight and SPY Buy & Hold respectively.

On the annualized basis – if we see we get around 20% return from our strategy while 9.3% and 10.9 % with Equal Weight & S&P Buy & Hold respectively.

The reward vs risk or Sharpe ratio is also impressive for our strategy compared to Equal Weight & SPY Buy & Hold thus giving us earn better return per unit of risk we take .

The most important result is however the max drawdown which significant smaller than SPY Buy & Hold. This is relevant and possible because of regime switches which provides protection during market stress.

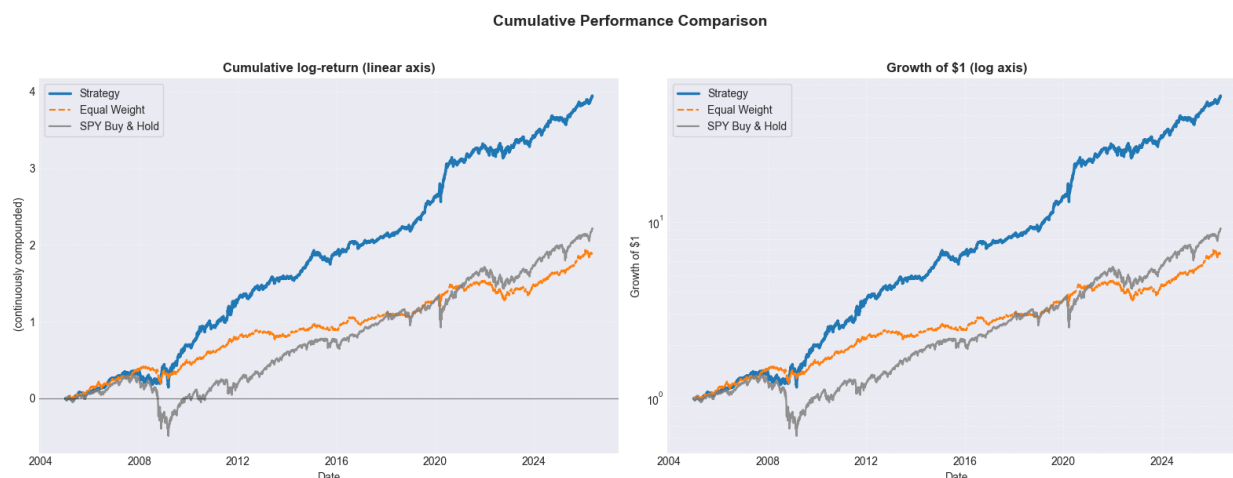


Fig10: Cumulative performance plot: Strategy vs equal weight vs SPY

The cumulative performance curve also shows that during crisis period as 2008 and 2020 the strategy shifts towards lesser riskier asset due to increased VIX regime

Table 13: Strategy Variant Table

Metric	Strategy (100% Best, Gross)	Strategy (100% Best, Net 5bps)	Strategy (60/40 Top-2, Gross)	Strategy (60/40 Top-2, Net 5bps)	Equal Weight	SPY Buy and Hold
Cumulative Return	50.1405	20.2016	25.3694	14.5445	5.6073	8.0825
CAGR	0.2026	0.1539	0.1658	0.1373	0.0926	0.1090
Volatility	0.1505	0.1505	0.1220	0.1220	0.0970	0.1899
Sharpe Ratio	1.3462	1.0226	1.3596	1.1256	0.9546	0.5739
Max Drawdown	-0.2599	-0.2688	-0.1609	-0.1785	-0.2295	-0.5519
Sortino Ratio	1.8469	1.4040	1.9124	1.5829	1.3002	0.6838
Calmar Ratio	0.7793	0.5728	1.0306	0.7689	0.4033	0.1975

Strategy enhancements:

We also extended our strategy with two spec:

1. 60/40 top two: here what we did we place the 60% to the ETF basket which gave us highest average return and remaining 40% went to the 2nd best higher average return
2. Transaction cost: here we charges transaction cost on the daily turnover to make it more realistic

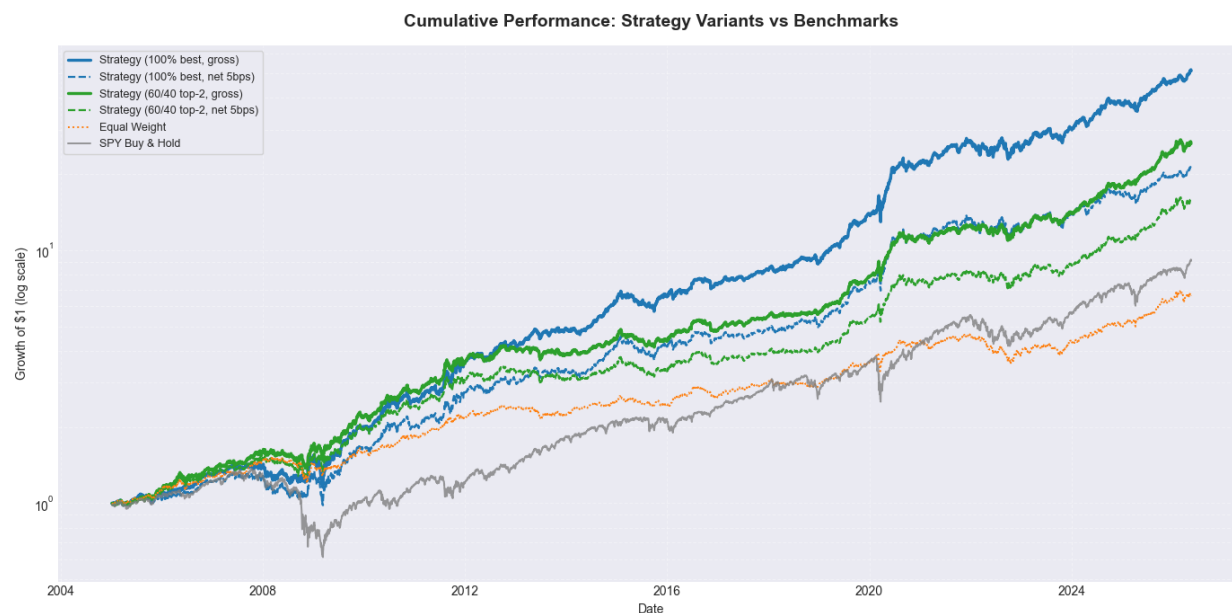


Fig11: Strategy Variant vs Benchmark

Strategy variant and transaction costs:

1. The transaction cost result should be interpreted carefully. The 5bps cost doesn't destroy the strategy but it has a meaningful impact. For 100 percent top-1 strategy- final amount falls from 50.14 x to 20.20x net, and CAGR falls from 20.26% to 15.39%.
2. As shown in 'Table 13: Strategy Variant Table' 60/40 top two underperforms 100% top one in CAGR but does have a slightly smoother path because of the reason diversification in the regime trades a very small return premium for standard drawdown reduction.

Sensitivity Analysis

The idea of this step is to understand if the results depend on one model too much, i.e. whether we are overfitting our analysis.

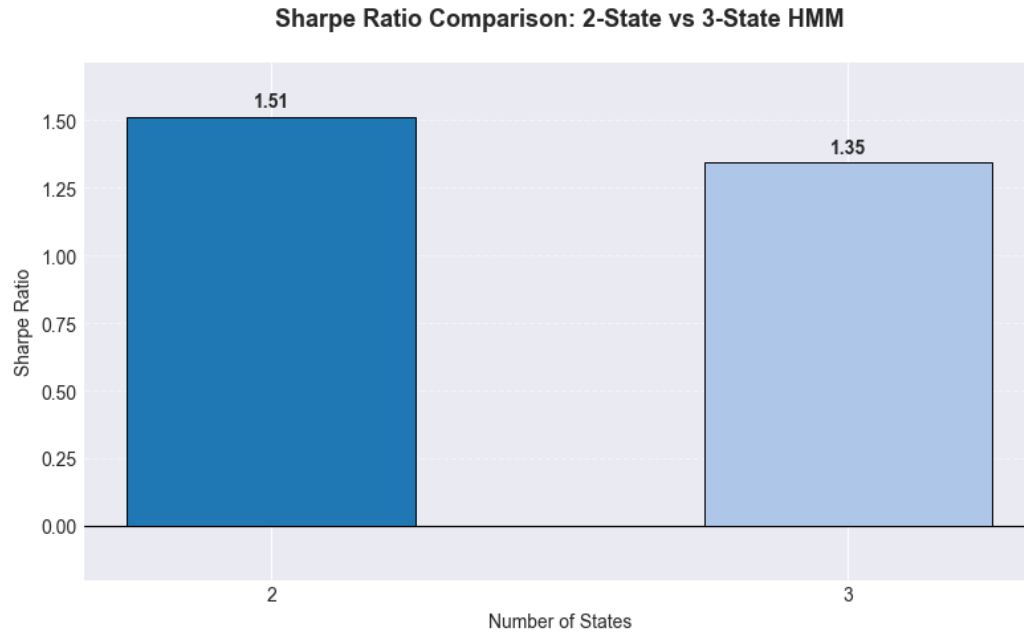


Fig12: 2 state vs 3 state Sharpe ratio comparison

Table 14: 2 state vs 3 state HMM comparison

States	Log Likelihood	AIC	BIC	Annual Return	Volatility	Sharpe Ratio	Max Drawdown
2	6879.5783	-13745.1565	-13699.0299	0.2310	0.1526	1.5141	-0.3042
3	6941.3508	-13854.7015	-13762.4483	0.2026	0.1505	1.3462	-0.2599

In the above table- we see both models perform well – but their metric have different strenght. 2-state has higher Sharpe to 3 state. However AIC & BIC are more economical for state 3 state. In this regard 3-HMM state becomes more preferable.

Table 15: Quantile threshold sensitivity for Markov chain sensitivity

Quantile Range	Annual Return	Volatility	Sharpe Ratio
0.33 - 0.66	0.0201	0.1691	0.1190
0.25 - 0.75	0.0382	0.1703	0.2242
0.20 - 0.80	0.0386	0.1733	0.2226

Based on the result for Sharpe ration for Markov chain for 3 quantile- we observe it performs weaker to that HMM model . This allows to support our decision to use HMM3 to simple quantile MCC model .

Table 16: 5 years rolling analysis for performance

Start Date	End Date	Annual Return	Volatility	Sharpe Ratio
2005-01-04	2010-01-05	0.0265	0.2348	0.1127
2006-01-04	2011-01-04	0.1145	0.1716	0.6673
2007-01-05	2012-01-04	-0.1314	0.2398	-0.5481
2008-01-07	2013-01-07	-0.0891	0.2513	-0.3545
2009-01-06	2014-01-07	0.0948	0.1746	0.5428
2010-01-06	2015-01-07	0.2324	0.1456	1.5962
2011-01-05	2016-01-07	-0.0135	0.1500	-0.0903
2012-01-05	2017-01-06	-0.0196	0.1260	-0.1555
2013-01-08	2018-01-08	-0.0061	0.1161	-0.0528
2014-01-08	2019-01-09	0.1818	0.1049	1.7324
2015-01-08	2020-01-09	0.1184	0.1272	0.9307
2016-01-08	2021-01-08	0.2800	0.1538	1.8205
2017-01-09	2022-01-07	0.3510	0.1532	2.2907
2018-01-09	2023-01-10	0.1940	0.1843	1.0523
2019-01-10	2024-01-11	0.1045	0.2024	0.5162
2020-01-10	2025-01-14	0.1777	0.1862	0.9540
2021-01-11	2026-01-15	0.0934	0.1601	0.5835

From above we see that strategy perform quite well in stress period ie around 2020 and 2022. It performs worse during quitter periods when we have less regime switches. This is kind of expected as regime rotation strategy adds most value when market conditions will change.

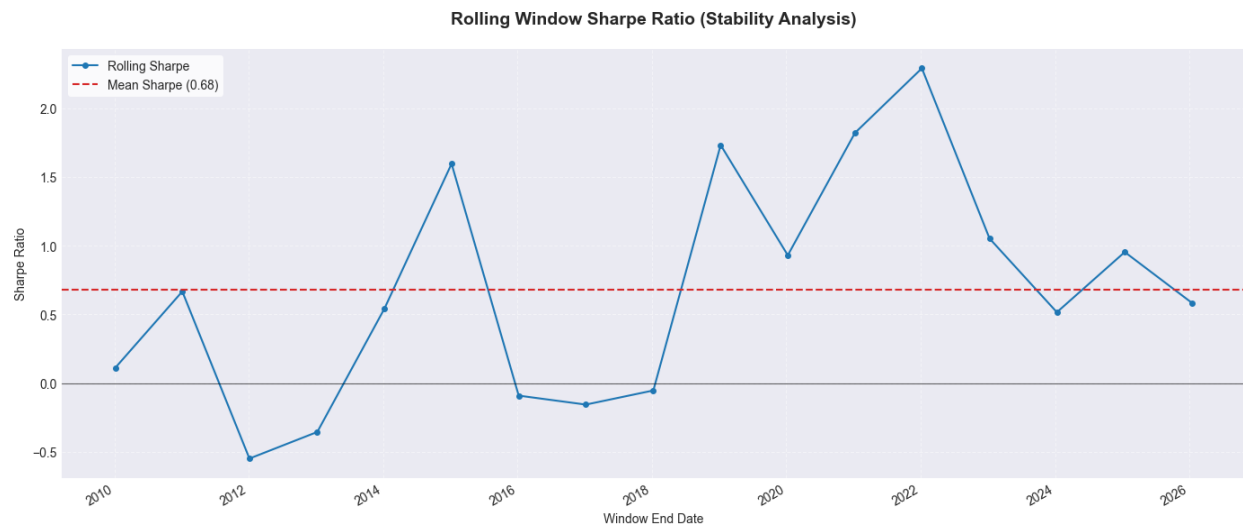


Fig13: Rolling window Sharpe analysis

Table 17: Q learning of regime labelled returns

Regime	Q-learning Policy	MDP Optimal Policy	Top-1 Heuristic	Q(s, SPY)	Q(s, TLT)	Q(s, GLD)
Stress (VIX spikes)	GLD	TLT	TLT	0.0697	0.0688	0.0784
Calm (low Delta VIX)	SPY	SPY	SPY	0.0809	0.0722	0.0721
Normal	TLT	TLT	TLT	0.0681	0.0740	0.0689

Q-learning[8] extension is useful as supports same idea- calm period prefer SPY, normal /stress defensive. One thing to note is that it choose GLD instead of TLT in stress state since it learns from noisy historical data. It also shows TLT and GLD are close defensive alternatives in stress period.

Conclusion

The above analysis provides useful information with respect to asset allocation by using volatility regimes. Using VIX log returns- HMM gives the best overall regime based on the metric on AIC, BIC and economic interpretation. The selected states we have created are interpreted as stress, calm and normal regimes.

Final allocation rule which we made is simple and transparent because it invests in SPY in calm period and TLT during stress. 1-day execution lag allows make the back test more realistic and avoid to have any look ahead bias.

Regime based strategy also outperforms both benchmark. The annualized return, Sharpe ratio, and lower drawdown in shown in above analysis table better than the SPY buy & hold. The equal weight benchmark do perform well as it is diversified but it does not match the strategy's Sharpe ratio performance.

Overall evidence suggest that our VIX based regime detection can help to improve portfolio allocation. The main benefit though is not better return but protection during the stressful. Hence it can be used to reduce equity exposure during high risk periods by rotating

Limitation of the model:

There are some limitation with the model.

First model depend only one parameter to identify regime ie VIX return. Though its useful it does not capture everything which impact the 3 ETFs. For eg. Other factor such as interest rates, inflation, monetary policy, geopolitical events etc also affect asset return

Second Markov & HMM approaches assume that the regime depends on transition matrix . This means that moving from one state regime to another there is associated probability , however as we know that crisis period such as 2008 financial crisis, Covid-19 in 2020 or high inflation phase cannot be determined statistically.

Third – The strategy is having sample biasness. The model is fitted using the historical return of the asset . Though we introduce 1-day lag still more realistic approach would have been to test with rolling or walk-forward approach which re-estimated using the data available only at that time.

Finally, model have transaction cost which is very simplified. Real trading involves bid-ask spreads, slippage, tax, execution delay and liquidity constraints. These realistic costs can lead to substantial performance impact on the strategy.

References

Course material (MScFE 622 – Stochastic Modelling, WorldQuant University)

1. *Module 5, Lesson 1 – Markov chains and reinforcement learning.* Markov property, transition matrices, -step probabilities.
2. *Module 5, Lesson 2 – Markov chains: credit-rating application.* Empirical transition matrices, stationary distribution, expected hitting times.
3. *Module 5, Lesson 3 – Hidden Markov Models.* Hamilton forward filter, predict/update recursions, log-likelihood.
4. *Module 5, Lesson 4 – EM algorithm & regime switching.* Baum–Welch updates for the Gaussian HMM; application to VIX regimes.
5. *Module 6, Lesson 1 – Markov Decision Processes.* MDP tuple, value and action-value functions, Bellman equations.
6. *Module 6, Lesson 2 – Policy iteration on gridworld.* Iterative policy evaluation and improvement.
7. *Module 6, Lesson 3 – Tactical asset allocation by RL.* 12-state SPY/IEF/SHY rotation, empirical and, 60/40 benchmark.
8. *Module 6, Lesson 4 – Q-learning.* Tabular Q-learning, -greedy exploration, temporal-difference target.

Standard references (methodology)

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22. Lebedev, S. *hmmlearn – Hidden Markov Models in Python with scikit-learn-like API*.
<https://github.com/hmmlearn/hmmlearn>
23. Yahoo! Finance – historical adjusted-close prices for SPY, TLT, GLD and ^VIX (accessed via yfinance, snapshot stored in market_data_close.csv).